MILITARY SPECIFICATION

LAUNCHERS, GRENADE, SMOKE: SCREENING, RP, M239 AND M250

This specification is approved for use by U.S. Army Chemical Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of launchers.

1.2 Classification. The launchers shall be furnished in the following types (see 6.3):

Type M239 - Launcher, Grenade, Smoke: Screening, RP, (TDPL13-12-32).
Type M250 - Launcher, Grenade, Smoke: Screening, RP, (TDPL13-12-122).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized and Containerized Unit Loads 40 Inch Pallets, Skids, Runners or Pallet Type Base.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research and Development Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 1040

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2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date Of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL13-12-32 - launcher, Grenade, Smoke: Screening, RP, M239.
TDPL13-12-122 - Launcher, Grenade, Smoke; Screening, RP, M250.

PUBLICATIONS: SPECIAL PACKAGING INSTRUCTION (SPI)

SPI P13-12-1.76 - Stowage Box, Smoke Grenade.

(Codes of standards drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text Of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL13-12-32 and TDPL13-12-122 shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the M239 and M250 launchers shall conform to the specifications and drawings listed on TDPL13-12-32 and TDPL-13-12-122.

3.2 Assembly. The M239 and M250 launchers shall be assembled as specified on Drawing 13-12-32 and 13-12-122.

3.3 Leakage.

3.3.1 Discharger. The assembled discharger, Drawing 13-12-34 and 13-12-35 shall not leak when subjected to an internal air pressure of 1.5 + .5 - 0.0 pounds per square inch gage (psig) (10.34 + 3.45 - 0.0 kilopascals) for a minimum period of 30 seconds, at ambient temperature when tested as specified in 4.4.4.1(a).

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3.3.2 Push button unit. The assembled unit, Drawing 13-12-37, shall show no evidence of leakage when tested as specified in 4.4.4.1(b).

3.4 Resistance and continuity.

3.4.1 Discharger resistance. The resistance measured between the points shall be in accordance with table I for right hand and left hand discharger when tested as specified in 4.4.4.2(a).

TABLE I. Resistance - discharger

FROM	TO	Resistance (ohms)
Receptacle connector Pin A	Contact pin of barrels 1, 2 and 5	#
	Contact pin of barrels 3, 4 and 6	15 ± 1.5
Receptacle connector Pin B	Adapter assemblies in all barrels	0
	Pin A (receptacle connector)	#
	Pin C	#
Receptacle connector Pin C	Contact pins of barrels 1, 2 and 5	15±1.5
	contact pins of barrels 3, 4 and 6	#

NOTES:
1. 0 indicates zero resistance.
2. # indicates infinite resistance.

3.4.2 Push button unit continuity. The continuity of the unit shall be in accordance with table II when tested as specified in 4.4.4.2(b).

3.5 Insulation resistance. The discharger and push button unit shall have an insulation resistance of not less than 10 megohms when tested as specified in 4.4.4.3.

3.6 Preproduction. Prior to the start of regular production, a production lot of launchers shall be produced (see 6.2) in accordance with this specification, for examination and tests (see 4.3).

3.7 Workmanship. The launcher shall be free from contamination (oil, grease. dirt. rust or chips) damage *or* deformation (dents, cracks, porosity) which could affect its intended use. The dischargers and push button unit shall comply with the applicable requirements of MIL-STD-454 for workmanship.

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TABLE II. Continuity - push button

Pin to pin	No switches pressed	Left switch pressed	Right switch pressed	Both switches pressed
AtoB	#	#	#	#
C	#	#	#	#
D	#	#	#	#
E	0	0	0	0
F	#	0	#	0
G	#	#	#	#
Btoc	#	#	0	0
D	0	0	0	0
E	#	#	#	#
F	#	#	#	#
G	#	#	#	#
ctoD	#	#	0	0
E	#	#	#	#
F	#	#	#	#
G	#	#	#	#
Dto.E	#	#	#	#
F	#	#	#	#
G	#	#	#	#
EtoF	#	0	#	0
G	#	#	#	#
FtoG	#	#	#	#

NOTES :

1. 0 indicates zero resistance.
2. # indicates infinite resistance.

40 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection

4.3.1 Lot. A preproduction lot of 5 launchers shall be produced using the same methods, materials, equipment and processes as will be used in regular production.

4.3.2 For examination and tests. The launchers shall be examined and tested in accordance with the classification of defects (4.4.3.2). The AQLs do not apply.

4.3.3 Acceptance/rejection criteria. The launchers shall comply with the examinations and test specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production

4.4 Quality conformance inspection,

4.4.1 Lotting. A lot shall consist of like components or subassemblies of one interfix number produced by one manufacturer, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling. Sampling shall be conducted in accordance with table III.

4.4.3 Inspection procedure.

4.4.3.1 For examination and test. Sample launchers and components shall be examined and tested in accordance with the classification of defects and table III.

4.4.3.2 For critical characteristics. Each discharger in the lot shall be tested for its critical characteristic listed in the classification of defects. Failure of any one discharger shall be cause for rejection of the lot represented.

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TABLE 111. Sampling and inspection

Requirement	Item	Sample size	AQL	Test
3.3.1	Discharger	MIL-STD-105	1.0	4.4.4.1(a)
3.4.1	Discharger	100$	---	4.4.4.2(a)
3.4,2	Push button unit	100%	---	4.4.4.2(b)
3.3.2	Push button unit	MIL-STD-105 level S-3	1.5	4.4.4.1(b)
3.5	Discharger	MIL-STD-105	1.0	4.4.4.3
3.5	Push button unit	MIL-STD-105	1.0	4.4.4.3

4.4.3.3 Classification of defects.

(a) Launcher, grenade, smoke: screening, RP, M239 anM250 (Drawing 13-12-32 and 13-12-122).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Item missing

(b) Discharger, smoke grenade (Drawing 13-12-34, left hand and 13-12-35, right hand).

Categories	Defects	Acceptance standards
Critical:		
1	Circuit resistance (see table 111)	4.4.4.2(a)

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Categories	Defects	Acceptance standards
Major:	AQL 1.0 percent defective	
101	Leakage	4.4.4.1(a)
102	Component missing or incorrectly assembled	
103	Soldering incorrect	
104	Terminal marking incorrect	
105	Protective finish missing or incorrectly applied	
106	Drain pipe incorrectly assembled	
107	Insulation resistance	4.4.4.3
108	Workmanship (3.7)	
109	Screw-type fasteners improperly torqued or loose	

(c) Discharger body (Drawing 13-12-51 left hand, and 13-12-52 right hand).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Basic dimensions incorrect
102	Barrel diameter incorrect
103	Barrel depth incorrect
104	Barrel wall thickness incorrect
105	Spot faces incorrect
106	Location of bosses incorrect
107	Hole locations incorrect
108	Hole sizes incorrect
109	Drain hole location incorrect
110	Surface finish incorrect
111	Protective finish missing or incorrect
112	Casting quality incorrect
113	Workmanship (3.7)

(d) Cover plate (Drawing 13-12-50).

Categories	Defects
Critical:	None defined
Major	AQL 1.0 percent defective
101	Hole locations incorrect
102	Hole diameters incorrect

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categories	Defects
103	Overall dimensions incorrect
104	Protective finish missing or incorrect
105	Workmanship (3.7)

(e) Drain pipe (Drawing 13-12-69 right hand and 13-12-68 left hand).

Categories	Defects
Critical:.	None defined
Major: AQL	1.0 percent defective
101	Dimensions incorrect
102	Workmanship (3.7)

(f) Nameplate (Drawing 13-12-67).

Categories	Defects
Critical:	None defined
Major: AQL	2.5 percent defective
101	Dimensions incorrect
102	Lettering incorrect
103	Hole locations incorrect
104	Color incorrect
105	Workmanship (3.7)

(g Push button unit (Drawing 13-12-37).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:		
101	Continuity incorrect (see table III)	4.4.4.2(b)
102	Component missing or incorrectly assembled	
103	Nameplate incorrect or incorrectly assembled	
104	Electrical connections loose	
105	Threads not coated before assembly	
106	Identification marking missing or incorrect	

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Categories	Defects	Acceptance standards
Major:	AQL 1.0 percent defective	
107	Insulation resistance	4.4.4.3
108	Workmanship (3.7)	
Minor	AQL 1.5 percent defective	
201	Leakage	4.4.4 .1(b)
202	Wiring diagram missing or incorrectly applied	
203	Protective finish missing or incorrect	

(h) Base plate (Drawing 13-12-76).

Categories	Defects
Critical:	None defined
Major:	AQL 2.0 percent defective
101	Hole locations incorrect
102	Hole diameters incorrect
103	Length incorrect
104	Width incorrect
105	Protective finish incorrect

(i) Harness (Drawing 13-12-79).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Lead length incorrect
102	Soldering incorrect
103	Heat shrink insulation missing or incorrect
104	Marking missing or incorrect
105	Component missing or incorrectly assembled

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(J) Push button box (Drawing 13-12-88).

Categories	Defects
Critical:	None defined
Major	AQL 1.5 percent defective
101	Dimensions (except as noted) incorrect
102	Hole locations incorrect
103	Spot faces incorrect
104	Threads incorrect
105	Surface finish incorrect
106	Protective finish missing or incorrect
107	Workmanship (3.7)

(k) Gasket (Drawing 13-12-89).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Hole locations incorrect
102	Hole diameters incorrect
103	Workmanship (3.7)

(1) Stowage box, smoke grenade (Drawing 13-12-176).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Pads missing, located incorrectly or loose
102	Catch nonfunctioning or loose
103	Protective finish missing or incorrect
Minor:	AQL 2.5 percent defective
201	Marking incorrect or improperly located
202	Workmanship (3.7)

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(m) Box (Drawing 13-12-177).

Categories		Defects
Critical:		None defined
Major	AQL 1.0 percent defective	
101		Length incorrect
102		Width incorrect
103		Height incorrect
104		Divider located incorrectly
105		Mounting lugs located incorrectly
106		Mounting lug holes incorrect
107		Hinges assembled incorrectly
Minor:	AQL 2.5 percent defective	
201		Welding incorrect
202		Protective finish missing or incorrect
203		Workmanship (3.7)

n) Packaging and packing (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Marking illegible or incorrect
102		Packaging materials incorrect
103		Dimensions incorrect
104		Loose pack
105		Box closure incorrect
106		Component missing
Minor:	AQL 4.0 percent defective	
201		Label smeared or not coated
202		Label characters size incorrect
203		Discharger incorrectly bolted to plywood panel

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage.

(a Discharger. The discharger, right hand and left hand, shall be subject to the specified pressure applied through the test port and submerged

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in water at ambient temperature to a maximum depth of 3 inches (76 millimeters) below the surface of the water for the required time. A continuous stream of bubbles indicates leakage.

(b) Push button unit.

(1) Method A. The unit shall be submerged in 3.0 ± 0.5 feet of water (914 40 milimeters at ambient temperature for 15 minutes. Upon removal, the unit shall be thoroughly dried and the base plate removed. Any water found inside the unit constitutes leakage.

(2) Method B. While submerged below the surface of water at ambient temperature and at a depth of 3 inches to 12 inches, the push button unit shall be subjected to 1.5 ± 0.5 psig air pressure through an acceptable test fixture which simulates the cover plate. A continuous stream of bubbles indicates leakage.

4.4.4.2 Resistance and continuity.

(a) Discharger resistance. The resistance shall be measured using an ohmmeter to determine the resistance between the pin (from) and the contact pin or body (to) as specified in 3.4.1.

(b) Push button unit continuity. The continuity shall be measured using an ohmmeter to determine the resistance between the pins of electrical connector. The resistance shall be as specified in 3.4.2.

4.4.4.3 Insulation resistance. The insulation resistance between pairs of component circuits and between each circuit and ground shall be as specified in 3.5 when measured at 500 volt direct current (vdc) minimum, for a minimum of 30 seconds.

5. PACKAGING

5.1 Preservation, level A.

5.1.1 Stowage box. The stowage box shall be packaged and marked as shown on Special Packaging Instruction P13-12-176 (M239 laucher).

5.1.2 Push button unit. The push button unit shall,be packaged and marked as shown on Drawing 13-12-137 (for M239 launcher).

5.1.3 Discharger. The discharger shall be packaged and marked as shown on Drawing 13-12-61 for M239 and M250 launchers.

5.1.4 Discharger cover. The discharger cover (1 left and 1 right) shall be packaged and marked as shown on Drawing 13-12-137 for M239 and 13-12-60 for M250 launcher.

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5.2 Packing, level A. Items packaged and marked as specified in 5.1, shall be packed together as shown on Drawing 13-12-185 for M239 launcher or 13-12-62 for M250 launcher.

5.3 Marking. Items packed as specified in 5.2, shall be subsequently marked for storage and shipment in accordance with drawings listed in 5.1 and 5.2.

5.4 Palletization. Units packed as specified on Drawing IS-I-2-62 shall be palletized in accordance with MIL-STD-147.

6. NOTES

6.1 Intended use. These items are intended for use as a vehicle mounted launcher for smoke grenades.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Preproduction:

(1) Time allowed for contractor submission of samples for any Government tests and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Type of launcher required.

6.3 Definitions.

(a) Launchers.

(1) The M239 consists of left and right dischargers, left and right covers, a push button unit and two stowage boxes.
(2) The M250 launcher consists of left and right dischargers and left and right covers.

(b) Dischargers. The dischargers are components of the launcher from which the smoke grenades are ejected.

6.4 Supersession data. This specification supersedes and includes the requirements of Chemical Systems Laboratory Purchase Description EA-L-1263A dated 29 October 1981.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A170

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

!. DOCUMENT NUMBER	20 DOCUMENT TITLE
MIL-L-51513 (EA)	LAUNCHERS, GRENADE, SMOKE: SCREENING, RP , M239 AND M250

3a NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, *City, State, ZIPCode)*

4. TYPE OF ORGANIZATION *f~C1'kOIU)*

n VENDOR

cl USER

• 1 **MANUFACTURER**

OTHER(Specfl'Y): "

5. PROBLEMAREAS

a Paragraph Number and Wording:

b. **Recommmended Wording:**

c. Reason/Rational for Recommendation:

5.-REMARKS

7a. NAME OF SUBMITTER *(Last, Firct,* **MI) -** Optionai	*b.* WORK TELEPHONE NUMBER (include code) - Optional
c. MAILING ADDRESS (Streef, *City,* State, *ZIP Code)* - Optionai	8. DATE OF SUBMISSION *(YyMhIDD)*

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